1. Find the 6th term in the expansion of $(1-2x^2)^{1/2}$:
NIMCET 2007
(a)$\frac{7}{8}x^5$ (b)$-\frac{7}{8}x^5$ (c)$\frac{7}{8}x^{10}$ (d)$\frac{-693}{32}x^{10}$

2. $(\sqrt{5}-2)^{1/3}-(\sqrt{5}+2)^{1/3}$ will be:
NIMCET 2007
(a)Irrational
(b)one
(c)rational but not integer
(d)none

3. $2C_0+\frac{2^2}{2}C_1+\frac{2^3}{3}C_2+\cdots+\frac{2^{11}}{11}C_{10}=$
NIMCET 2007
(a)$\frac{3^{11}-1}{11}$ (b)$\frac{2^{11}-1}{11}$
(c)$\frac{11^3-1}{11}$ (d)$\frac{11^2-1}{11}$

4. If $(1+x-2x^2)^6=1+a_1x+a_2x^2+\cdots+a_{12}x^{12}$, then the value of $a_2+a_4+\cdots+a_{12}$ is
NIMCET 2009
(a)1024 (b)64
(c)32 (d)31

5. If $(1+x)^n=a_0+a_1x+a_2x^2+\cdots+a_nx^n$, then $\left(1+\frac{a_1}{a_0}\right)\left(1+\frac{a_2}{a_1}\right)\left(1+\frac{a_3}{a_2}\right)+\cdots+\left(1+\frac{a_n}{a_{n-1}}\right)$ is equal to
NIMCET 2010
(a)$\frac{n^n}{n!}$ (b)$\frac{(n+1)^n}{n!}$
(c)$\frac{n^{n+1}}{(n+1)!}$ (d)$\frac{(n-1)^n}{n!}$

6. The sum of ${}^{20}C_8+{}^{20}C_9+{}^{21}C_{10}+{}^{22}C_{11}-{}^{23}C_{11}$ is
NIMCET 2012
(a) ${}^{22}C_{12}$ (b) ${}^{23}C_{12}$
(c)0 (d) ${}^{21}C_{10}$

7. If n and r are integers such that $1\le r\le n$, then the value of $n({}^{n-1}C_{r-1})$ is
NIMCET 2014
(a) ${}^{n}C_r$ (b)$r({}^{n}C_r)$
(c)$n({}^{n}C_r)$ (d)$(n-1)({}^{n}C_r)$

8. If x is so small that x^2 and higher powers of x can be neglected, then $\frac{(9+2x)^{\frac{1}{2}}(3+4x)}{(1-x)^{1/5}}$ is approximately equal to
NIMCET 2014
(a)$9+\frac{74}{15}x$ (b)$9+\frac{74x}{5}$
(c)$3+\frac{74}{15}x$ (d)$3+\frac{74}{5}x$

9. If $(1-x+x^2)^n=a_0+a_1x+a_2x^2+\cdots+a_{2n}x^{2n}$, then $a_0+a_2+a_4+\cdots+a_{2n}$ is
NIMCET 2017
(a)$\frac{3^n+1}{2}$ (b)$\frac{3^n-1}{2}$ (c)$\frac{1-3^n}{2}$ (d)$3^n+\frac{1}{2}$

10. The coefficient of x^n in the expansion of $(1-2x+3x^2-4x^3+\cdots to\ \infty)^{-n}$ is
NIMCET 2018
(a)$\frac{(2n)!}{n!(n-1)}$ (b)$\frac{(2n)!}{[(n-1)!]^2}$
(c)$\frac{(2n)!}{(n!)^2}$ (d)None of these

11. If $(1+x-2x^2)^6=1+a_1x+a_2x^2+\cdots a_{12}x^{12}$, then the value of $a_2+a_4+a_6+\cdots+a_{12}$ is
NIMCET 2019
(a)29 (b)30
(c)31 (d)32

12. If a and b are the greatest value of ${}^{2n}C_r$ and ${}^{(2n-1)}C_r$ respectively, then
NIMCET 2019
(a)$a=2b$ (b)$b=2a$
(c)$a=b$ (d)$a^2=2b^2$

13. If $\binom{15}{8}+\binom{15}{7}=\binom{n}{r}$, then the value of n and r are
NIMCET 2020
(a)16 and 7 (b)16 and 8
(c)16 and 9 (d)30 and 15

14. If $\log(1-x+x^2)=a_1x+a_2x^2+a_3x^3+$ ......Then, $a_3+a_6+a_9+$..... equal to
NIMCET 2021
(a)$\log 2$ (b)$\frac{1}{3}\log 2$
(c)$\frac{2}{3}\log 2$ (d)$2\log 2$

15. If $\frac{n!}{2!(n-2)!}$ and $\frac{n!}{4!(n-4)!}$ are in ratio 2 : 1, then the value of n is
NIMCET 2021
(a)0 (b)2
(c)4 (d)5

16. If n is an integer between 0 to 21, then find a value of n for which the value of $n!(21-n)!$ is minimum.
NIMCET 2021
(a)9 (b)10 (c)12 (d)21

17. If $x=1+\sqrt[6]{2}+\sqrt[6]{4}+\sqrt[6]{8}+\sqrt[6]{16}+\sqrt[6]{32}$, then $\left(1+\frac{1}{x}\right)^{24}=$ (Binomial)
(a) 1 (b) 4
(c) 16 (d) 24
NIMCET 2024

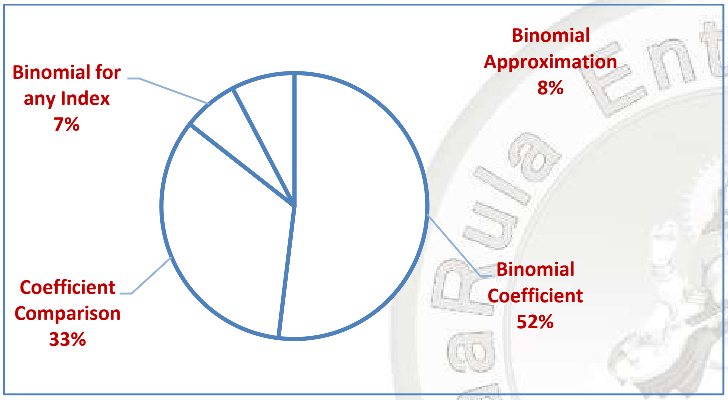
ANSWER KEY

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
d	b	a	d	b	c	b	b	a	c
11.	**12.**	**13.**	**14.**	**15.**	**16.**	**17.**			
c	a	b	c	d	b	c			

Analysis 2007 to 2025
Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	03	NIMCET 2016	00
NIMCET 2008	00	NIMCET 2017	01
NIMCET 2009	01	NIMCET 2018	01
NIMCET 2010	01	NIMCET 2019	02
NIMCET 2011	00	NIMCET 2020	01
NIMCET 2012	01	NIMCET 2021	03
NIMCET 2013	00	NIMCET 2022	00
NIMCET 2014	02	NIMCET 2023	00
NIMCET 2015	00	NIMCET 2024	01
		NIMCET 2025	00

Sub Topicwise Analysis



Sub topicwise Questions Asked in NIMCET

Sub topic	Question Asked
Binomial Coefficient	07
Coefficient Comparison	04
Binomial for any Index	01
Binomial Approximation	01

SOLUTION

1. **(d)** 6^{th} term in the exp. $(1-2x^2)^{1/2}$

$F_r = (1-x)^n \to r+1$ th term

$$T_{r+1} = \frac{n(n-1)(n-2)\cdots(n-\gamma+1)}{\gamma!}(x)^\gamma$$

$$T_{s+1} = \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)\left(\frac{1}{2}-3\right)\left(\frac{1}{2}-4\right)\left(\frac{1}{2}-5\right)\left(\frac{1}{2}-6\right)\times(-2x)^5}{5!}$$

$$= \frac{\frac{1}{2}\times\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)\left(-\frac{7}{2}\right)\left(-\frac{9}{2}\right)\times\left(-\frac{11}{2}\right)\cdot(2)^5\times x^{10}}{120}$$

$$= -\frac{1}{2^2}\times\frac{2^5\times108\times5\times7\times9\times11}{120}$$

$$= \frac{-7\times9\times11}{2^5}\cdot x^{10^{408}}$$

$$\Rightarrow -\frac{693}{2^5}x^{10}$$

2. **(b)**

3. **(a)**

sum of $2C_0 + \frac{C_1}{2}2^2 + \frac{C_2}{3}2^3 + \cdots - \frac{C_n}{10+1}2^{n+1}$

$(1+x)^n = {}^nc_1 + {}^nc_1x + {}^nc_2x^2 + \cdots {}^nc_nx^n$

$\int (1+x)^n = \int ({}^nc_1 + {}^nc_1x + {}^nc_2x^2 + \cdots {}^nc_nx^n$

$$\frac{(1+x)^{n+1}}{n+1} = c_0x + c_1\frac{x^2}{2} + c_2\frac{x^3}{3} + \cdots c_n\frac{x^{n+1}}{n+1} + c$$

Put x = 0

$\frac{1+0}{n+1} = 0 + 0.....+ C$

C = $\frac{1}{n+1}$

$$6x + \frac{c_1x^2}{2} + c_2\frac{x^3}{3}\cdots\frac{l_nx^{n+2}}{n+1} = \frac{(1+x)^{n+1}}{n+1} - \frac{1}{n+1}$$

$x = 2\ n = 10$

$$2C_0 + \frac{2^2c_1}{2} + \frac{2^3}{3}c_2\cdots\cdot\frac{c_{10}}{10+1}\times 2^{11} = \frac{(1+2)^{11}}{11} - \frac{1}{11}$$

$$= \frac{3^{11}-1}{11}$$

4. **(d)** $(1+x-2x^2)^6 = 1 + a_1x + a_2x^2 + \cdots + a_{12}x^{12}$

Put $x = 1$

$(1+1-2)^6 = 1 + a_1 + a_2 + a_3 + \cdots + a_{12}$

Put $x = -1$

$(1-1-2)^6 = 1 - a_1 + a_2 - a_3 + a_4 \ldots + a_{12}$

$\Rightarrow (-2)^6 = 1 - a_1 + a_2 - a_3 + a_4 \ldots + a_{12}$

Add Eqs. (i) and (ii)

$0 + (-2)^6 = 2 + 2(a_2 + a_4 + a_6 + \cdots + a_{12})$

$\Rightarrow \frac{64-2}{2} = a_2 + a_4 + a_6 + \cdots + a_{12}$

$\Rightarrow 31 = a_2 + a_4 + a_6 + \cdots + a_{12}$

5. **(b)**

$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \cdots + {}^nC_nx^n$

Given, $(1+x)^n = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_nx^n$

$a_2 = {}^nC_2 = \frac{n(n-1)}{2!} \Rightarrow a_3 = {}^nC_3 = \frac{n(n-1)(n-2)}{3!}$

Otherwise use $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$

$\therefore \left(1+\frac{a_1}{a_0}\right)\left(1+\frac{a_2}{a_1}\right)\left(1+\frac{a_3}{a_2}\right)\ldots\left(1+\frac{a_n}{a_{n-1}}\right)$

$= \left(1+\frac{n}{1}\right)\left[1+\frac{n(n-1)}{2(n)}\right]\left[1+\frac{n(n-1)(n-2)(2)}{3!n(n-1)}\right]\ldots\left[1+\frac{1}{n}\right]$

$= [1+n]\left[1+\frac{n-1}{2}\right]\left[1+\frac{n-2}{3}\right]\ldots\left[1+\frac{1}{n}\right]$

$= \frac{(1+n)}{1}\frac{(n+1)}{2}\frac{(n+1)}{3}\frac{(n+1)}{4}\ldots\frac{(n+1)}{n} = \frac{(n+1)^n}{n!}$

6. **(c)** $({}^{20}C_9 + {}^{20}C_8) + {}^{21}C_{10} + {}^{22}C_{11} - {}^{23}C_{11}$

$= ({}^{21}C_9 + {}^{21}C_{10}) + {}^{22}C_{11} - {}^{23}C_{11}$

$[\because {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r]$

$({}^{22}C_{10} + {}^{22}C_{11}) - {}^{23}C_{11} = {}^{23}C_{11} - {}^{23}C_{11} = 0$

7. **(b)** $n({}^{n-1}C_{r-1}) = \frac{n\cdot(n-1)!}{(n-r)!(r-1)!} = \frac{n!}{(n-r)!(r-1)!}\times\frac{r}{r}$

$= \frac{r\cdot(n)!}{(n-r)!r(r-1)!} = \frac{(r)\cdot(n!)}{(n-r)!r!} = r({}^nC_r)$

8. **(b)**

Using Binomial approximation $(1+x)^n \cong 1+nx$

$\frac{(9+2x)^{1/2}(3+4x)}{(1-x)^{1/5}} = (9+2x)^{1/2}(3+4x)(1-x)^{-1/5}$

$= 9^{1/2}\left(1+\frac{2x}{9}\right)^{1/2}(3+4x)(1-x)^{-1/5}$

$= 3\left(1+\frac{1}{2}\times\frac{2x}{9}\right)3\left(1+\frac{4x}{3}\right)\left(1-\left(-\frac{1}{5}\right)x\right)$

$= 9\left(1+\frac{x}{9}\right)\left(1+\frac{4x}{3}\right)\left(1+\frac{x}{5}\right) = 9\left[1+\frac{4x}{3}+\frac{x}{9}\right]\left(1+\frac{x}{5}\right)$

$= 9\left[1+\frac{13x}{9}\right]\left(1+\frac{x}{5}\right) = 9\left[1+\frac{x}{5}+\frac{13x}{9}\right]$

$= 9\left[1+\frac{74x}{45}\right] = 9+\frac{74x}{5}$

9. (a) $(1-x+x^2)^n = a_0 + a_1x + a_2x^2 + \cdots + a_{2n}x^{2n}$

Put $x=1$

$(1-1+1)^n = a_0 + a_1 + a_2 + a_3 + \cdots + a_{2n}$

$\Rightarrow 1 = a_0 + a_1 + a_2 + a_3 + \cdots + a_{2n}$

Put $x=-1$

$[1-(-1)+(-1)^2]^n = a_0 - a_1 + a_2 - a_3 + a_4 \ldots + a_{2n}$

$\Rightarrow 3^n = a_0 - a_1 + a_2 - a_3 + a_4 + \cdots + a_{2n} \ldots$

Add Eqs. (i) and (ii),

$3^n + 1 = 2(a_0 + a_2 + a_4 + a_6 + \cdots + a_{2n})$

$\Rightarrow \frac{3^n+1}{2} = a_0 + a_2 + a_4 + a_6 + \cdots + a_{2n}$

10. (c) We should know,

$(1-2x+3x^2-4x^3+\cdots\infty) = (1+x)^{-2}$

$\Rightarrow (1-2x+3x^2-4x^3+\cdots\infty)^{-n}$

$[(1+x)^{-2}]^{-n} = (1+x)^{2n}$

General term of $(1+x)^{2n}$ is

$T_{r+1} = {}^{2n}C_r x^r$

For coefficient of x^n

$r = n = {}^{2n}C_n = \frac{(2n)!}{(2n-n)!n!} = \frac{(2n)!}{(n!)(n!)}$

$= \frac{(2n)!}{(n!)^2}$

11. (c) Refer to solution 1.

12. (a) Greatest value of ${}^nC_r$ is ${}^nC_{n/2} \rightarrow$ if n is even, and

${}^nC_{\frac{n-1}{2}}$ or ${}^nC_{\frac{n+1}{2}} \rightarrow$ if n is odd

$\therefore 2n$ is even

$\Rightarrow a = {}^{2n}C_{\frac{2n}{2}} = {}^{2n}C_n$

$\Rightarrow b = {}^{2n-1}C_{\frac{2n-1-1}{2}} = {}^{2n-1}C_{n-1}$

On dividing Eq. (i) by Eq. (ii), we get

$\frac{a}{b} = \frac{{}^{2n}C_n}{{}^{(2n-1)}C_{n-1}} = = \frac{(2n)!(n-1)!n!}{n!n!(2n-1)!} = \frac{2n(2n-1)!(n-1)!}{n(n-1)!(2n-1)!}$

$\frac{a}{b} = 2 \Rightarrow a = 2b$

13. (b)

${}^{15}C_8 + {}^{15}C_7 = {}^nC_r$

$\Rightarrow {}^{15+1}C_8 = {}^nC_r$

$\Rightarrow {}^{16}C_8 = {}^nC_r$

$\Rightarrow$ n = 16 and $r = 8$

14. (c) Given, $\log(1-x+x^2) = a_1x + a_2x^2 + a_3x^3 + \cdots$

Here, we use concept of cube roots of unity.

$1+\omega+\omega^2 = 0$ and $\omega^3 = 1$

Put $x=1$ in the Eq. (i),

$\log(1-1+1) = a_1 + a_2 + a_3 + \ldots\ldots..$

$\Rightarrow \log 1 = a_1 + a_2 + a_3 + a_4 + \cdots$

Put $x=\omega$ in Eq. (i)....

$\log(1-\omega+\omega^2) = a_1\omega + a_2\omega^2 + a_3\omega^3 + a_4\omega^4 + \ldots\ldots..$

$\Rightarrow \log(1-\omega+\omega^2) = a_1\omega + a_2\omega^2 + a_3\omega^3 + a_4\omega^3\omega + \cdots$

$\Rightarrow \log(1-\omega+\omega^2) = a_1\omega + a_2\omega^2 + a_3 + a_4\omega + \cdots$ $[\because \omega^3 = 1]$

$\Rightarrow \log(1-\omega+\omega^2) = a_1\omega + a_2\omega^2 + a_3 + a_4\omega + \cdots \ldots$ (iii)

Put $x=\omega^2$ in Eq. (i)

$\log(1-\omega^2+\omega^4) = a_1\omega^2 + a_2\omega^4 + a_3\omega^6 + \cdots$

$\Rightarrow \log(1-\omega^2+\omega) = a_1\omega^2 + a_2\omega + a_3 + \cdots$

Add Eqs. (ii), (iii) and (iv),

$\log 1 + \log(1-\omega+\omega^2) + \log(1-\omega^2+\omega)$

$= a_1(1+\omega+\omega^2) + a_2(1+\omega+\omega^2)$

$+3a_3 + a_4(1+\omega+\omega^2) + \cdots$

$\Rightarrow 0 + \log[(1+\omega^2)-\omega] + \log[(1+\omega)-\omega^2]$

$= 3[a_3 + a_6 + a_9 \ldots]$

$\Rightarrow \log[-\omega-\omega] + \log[-\omega^2-\omega^2]$

$= 3[a_3 + a_6 + a_9 \ldots] \left\{\begin{array}{l}\because 1+\omega+\omega^2 = 0 \\ \Rightarrow 1+\omega = -\omega^2 \\ \text{or } 1+\omega^2 = -\omega\end{array}\right\}$

$\Rightarrow \log[-2\omega] + \log[-2\omega^2] = 3[a_3 + a_6 + a_9 + \cdots]$

$\Rightarrow \log[(-2\omega)(-2\omega^2)] = 3[a_3 + a_6 + a_9 + \cdots]$

$\Rightarrow [\because \log a + \log b = \log ab]$

$\Rightarrow \log[4\omega^3] = 3[a_3 + a_6 + a_9 + \cdots]$

$\Rightarrow \frac{2\log 2}{3} = a_3 + a_6 + a_9 + \cdots$

15. (d)

$\frac{\frac{n!}{2!(n-2)!}}{\frac{n!}{4!(n-4)!}} = \frac{2}{1}$

$\Rightarrow \frac{n!4!(n-4)!}{2!(n-2)!n!} = \frac{2}{1}$

$\Rightarrow \frac{(n-4)!4\times3\times2!}{(n-2)(n-3)(n-4)!2!} = \frac{2}{1}$

$\Rightarrow \frac{12}{(n-2)(n-3)} = \frac{2}{1}$

$\Rightarrow 6 = (n-2)(n-3)$

$\Rightarrow n^2 - 5n + 6 = 6 \Rightarrow n(n-5) = 0$

$\Rightarrow n = 0,5$

n can not be zero because $(n-2)!$ cannot be defined

$\therefore n = 5$

16. (b) $n!(21-n)! = \frac{n!(21-n)!}{(21)!} \times (21)$

[Multiply and divide by (21)!]

$= \frac{(21)!}{\frac{(21)!}{n!(21-n)!}} = \frac{(21)!}{{}^{21}C_n} \left\{\because \frac{(21)!}{n!(21-n)!} = {}^{21}C_n\right\}$

For $\frac{(21)!}{{}^{21}C_n}$ to be minimum, ${}^{21}C_n$ has to be maximum

$\Rightarrow$ Maximum value of ${}^{21}C_n = {}^{21}C_{\frac{21-1}{2}} = {}^{21}C_{10}$

$\Rightarrow n = 10 \left\{ \begin{array}{l} \because 21 \text{ is odd} \\ \therefore \text{ Greatest term } = {}^{21}C_{21-1/2} \end{array} \right\}$

17. (c)

x = 1 + $2^{1/6}$ + $2^{2/6}$ + $2^{3/6}$ + ...+ $2^{5/6}$

a =1, r = $2^{1/6}$

$S_{S=} \frac{1.\left(1-\left(2^{1/6}\right)^6\right)}{1-2^{1/6}} = \frac{-1}{1-2^{1/6}} = \frac{1}{2^{2/6}-1}$

Value of $\left(1+\frac{1}{x}\right)^{24}$ = 2^4 = 16